

QPL 2026 program (17 – 21 August 2026 in Amsterdam, NL)



Overview of the program

Welcome to Amsterdam! Registration opens at 8:50 AM on Monday, August 17, at the University of Amsterdam's Roeterseiland campus. The conference runs through Friday, August 21, 2026, with three parallel sessions each afternoon. On Wednesday, there is a boat tour and conference dinner after the parallel sessions. On Thursday, a career fair follows the parallel sessions.

	Monday, August 17th	Tuesday, August 18th	Wednesday, August 19th	Thursday, August 20th	Friday, August 21st
8:50 – 9:30	Registration	×	×	×	×
9:30 – 10:15	Long Plenary Talks	Long Plenary Talks	Long Plenary Talks	Long Plenary Talks	Long Plenary Talks
10:15 – 11:00					
11:00 – 11:30	Coffee Break				
11:30 – 11:55	Short Plenary Talks	Short Plenary Talks	Industry Session	Short Plenary Talk	Business Meeting
11:55 – 12:20				Conference Photo	
12:30 – 14:30	Lunch Break				
14:30 – 14:55	Parallel Sessions	Parallel Sessions	Parallel Sessions	Parallel Sessions 14:30 – 16:10	Parallel Sessions
14:55 – 15:20					
15:20 – 15:45					
15:45 – 16:15	Coffee Break				
16:15 – 16:40	Parallel Sessions	Parallel Sessions	Parallel Sessions 16:15 – 17:05	Coffee Break	Parallel Sessions 16:15 – 17:05
16:40 – 17:05					
17:05 – 17:30					×
	×	Poster Session 17:30 – 19:30	Boat Tour 17:30 – 18:30		
			Conference Dinner 19:00 onwards		

Monday, August 17th

8:50 – 9:30	Registration
Long plenary talks	
9:30 – 10:15	Alex Maltesson, Ludvig Rodung, Niklas Budinger, Giulia Ferrini, Cameron Calcluth Equivalence of continuous- and discrete-variable gate-based quantum computers with finite energy
10:15 – 11:00	Raphaël Mothe, Jessica Bavaresco Efficient quantum-circuit simulation of classical control of causal order
Coffee break	
Short plenary talks	
11:30 – 11:55	Vinicius Pretti Rossi, Beata Zjawin, Roberto D. Baldijão, David Schmid, John H. Selby, Ana Belén Sainz How typical is contextuality?
11:55 – 12:20	Dichuan Gao, Razin A. Shaikh, Aleks Kissinger Graphical algebraic geometry: from ideals and varieties to qudit ZH completeness
Lunch break	

	Parallel A	Parallel B	Parallel C
14:30 – 14:55	Yilè Ying, Maria Ciudad Alanon, Daniel Centeno, Jacopo Surace, Marina Maciel Ansanelli, Ruizhi Liu, David Schmid, Robert Spekkens On whether quantum theory needs complex numbers: the foil theories perspective	Tomoaki Kawano, Ryo Kashima Restricted negation in orthomodular logic	Alexandre Clément A complete equational theory for real-Clifford+CH quantum circuits
14:55 – 15:20	Timothée Hoffreumon, Mischa P. Woods On the experimental falsification of real QT	Alexandru Baltag, Sonja Smets Logic meets Wigner's friend (and their friends)	Xiaoning Bian, Sarah Meng Li, Neil J. Ross, John van de Wetering, Yuming Zhao A complete and natural rule set for multi-qudit Clifford circuits in all odd prime dimensions
15:20 – 15:45	Timothée Hoffreumon, Mischa P. Woods A real matrix theory consistent with QT	Bert Lindenhovius, Vladimir Zamdzhiev Operator spaces, linear logic and the Heisenberg–Schrödinger duality of quantum theory	Colin Blake Completeness for prime-dimensional phase-affine circuits
Coffee break			
16:15 – 16:40	Beata Zjawin, Marina Maciel Ansanelli, David Schmid, Yilè Ying, John H. Selby, Ciarán M. Gilligan-Lee, Ana Belén Sainz, Robert Spekkens The resource theory of causal influence and knowledge of causal influence	Priyaa Varshinee Srinivasan, Jean-Simon Pacaud Lemay, Robin Cockett Generalized inverses of quantum channels: a categorical perspective	Fedor Kuyanov, Aleks Kissinger Efficient classical simulation of low-rank-width quantum circuits using ZX-calculus

16:40 – 17:05	Leonardo Vaglini, Nasra Daher Ahmed, Ravi Kunjwal Causal inequalities witness non-stabilizerness without magic	Andre Kornell, Bert Lindhovius The category of quantum graphs is closed	Kwok Ho Wan, Zhenghao Zhong, Ainhoa Zapirain Simulating magic state cultivation with few Clifford terms
17:05 – 17:30	Carla Ferradini, Giulia Mazzola, V. Vilasini Emergent causal order and time direction: bridging causal models and tensor networks	James Hefford Nuclearity and trace in monoidal bicategories with application to extended CFTs	Mark Koch Classical Clifford+T sampling without computing marginals

Tuesday, August 18th

Long plenary talks	
9:30 – 10:15	Maximilian Rüscher, Aleks Kissinger, Benjamin Rodatz / Benjamin Rodatz, Boldizsár Poór, Aleks Kissinger Completeness for fault equivalence of Clifford ZX diagrams / Fault tolerance by construction
10:15 – 11:00	Samson Abramsky, Rui Soares Barbosa, Carmen Constantin, Martti Karvonen Algebraic paradoxes in adaptive quantum computation
Coffee break	
Short plenary talks	
11:30 – 11:55	Manuel Mekkonen, Thomas D. Galley, Markus P. Müller Invariance under quantum permutations rules out parastatistics
11:55 – 12:20	Marek Arsenault, Hlér Kristjánsson A higher-order perspective on quantum signal processing
Lunch break	

	Parallel A	Parallel B	Parallel C
14:30 – 14:55	Andrey Boris Khesin, Sarah Meng Li, Boldizsár Poór, Benjamin Rodatz, John van de Wetering, Richie Yeung SpiderCat: optimal fault-tolerant cat state preparation	Roberto D. Baldijão, Marco Erba, David Schmid, John Selby, Ana Belén Sainz Tomographically-nonlocal entanglement	Matthew Wilson Agent policies from higher-order causal functions
14:55 – 15:20	Kwok Ho Wan, Henry Price, Qing Yao Holographic codes seen through ZX-calculus	Tim Achenbach, Andreas Bluhm, Leevi Leppäjärvi, Ion Nechita, Martin Plávala Factorization of multimeters: a unified view on nonclassical quantum phenomena	V. Vilasini, Lin-Qing Chen, Liuhang Ye, Renato Renner Events and their localisation are relative to a lab
15:20 – 15:45	Cole Comfort, Giovanni de Felice The delayed stabiliser ZX-calculus	Cihan Okay, Aziz Khoroof The geometry of fiber products of probability polytopes	Zixuan Liu, Ognian Oreshkov Parity erasure: a foundational principle for indefinite causal order
Coffee break			
16:15 – 16:40	Vincenzo Fiorentino, Kuntal Sengupta Superposition and its connections to uncertainty, entanglement and the quantum tensor product	Tom Williams, Mina Doosti, Farid Shahandeh Sheaf-theoretic preparation contextuality via stochastic extension	Luca Apadula, Alexei Grinbaum, Časlav Brukner Reference frames for process matrices: from coordinate parametrization to spacetime representation
16:40 – 17:05	Gaurang Agrawal, Matt Wilson Deriving the generalised Born rule from first principles	Theodoros Yianni, Farid Shahandeh, Nyan Raess Linear algebra of generalized contextuality in prepare-transform-measure scenarios	Raphaël Le Bihan, Alastair Abbott, Mnacho Echenim Probing the composition of processes with first-order-ISOMIX logic

17:05 – 17:30	James Hefford, Matt Wilson Quantum theory can decohere from a causally-indefinite post-quantum theory	David Schmid, Roberto D. Baldijão, John Selby, Ana Belen Sainz, Robert W. Spekkens Noncontextuality inequalities for prepare-transform-measure scenarios	Yassine Benhaj, Kuntal Sengupta, Cyril Branciard How many systems can be dephased before the quantum switch becomes causally definite?
Poster session (with reception) 17:30 – 19:30			

Wednesday, August 19th

Long plenary talks	
9:30 – 10:15	Miriam Backens, Simon Perdrix / Miriam Backens Completeness for flow-preserving rewrite rules / Generating one-way computations with flow: flow-preserving rewriting that ignores the interpretation
10:15 – 11:00	Quanlong Wang, Richard D. P. East, Razin A. Shaikh, Lia Yeh, Boldizsár Poór, Bob Coecke Beyond Penrose tensor diagrams with the ZX calculus: applications to quantum computing, quantum machine learning, condensed matter physics, and quantum gravity
Coffee break	
Industry session 11:30 – 12:20	
Lunch break	

	Parallel A	Parallel B	Parallel C
14:30 – 14:55	Luca Apadula, Alessandro Bisio, Giulio Chiribella, Paolo Perinotti, Kyrlo Simonov Higher-order transformations of bidirectional quantum processes	Nathan Claudet, Simon Perdrix Insights in graph state entanglement via r-local complementation: structure and a quasi-polynomial algorithm	Pablo Arrighi, Doğukan Bakircioglu, Nathan Houyet Quantum theory over dual-complex numbers
14:55 – 15:20	Matthew Wilson, James Hefford, Timothée Hoffreumon Supermaps on generalised theories	Piotr Mitosek, Miriam Backens Working with measurement-based computations on qudits	Nicolas Moulounguet, Augustin Vanrietvelde Subsystems as subsets of quantum channels, and the strange case of blind agents
15:20 – 15:45	Samson Abramsky, Radha Jagadeesan Essential unitarity for higher-order quantum computation	Aleks Kissinger, John van de Wetering ZX-flow: a flexible criterion for deterministic computation with ZX-diagrams	John Harding, Alexander Wilce Classical explanations in (and of) general probabilistic theories
Coffee break			
16:15 – 16:40	Thomas Bartsch, Yuhan Gai, Sakura Schäfer-Nameki Beyond Wigner – how non-invertible symmetries preserve probabilities	Haytham McDowall-Rose, Razin A. Shaikh, Lia Yeh From fermions to qubits: a ZX-calculus perspective	Amrapali Sen, Flavio Del Santo Superluminal transformations and indeterminism
16:40 – 17:05	Vanessa Brzić, Satoshi Yoshida, Mio Murao, Marco Túlio Quintino Higher-order quantum computing with known input states	Lia Yeh, Jiaxin Huang, Aleks Kissinger, Sarah Meng Li, John van de Wetering A three-way normal form for stabiliser codes across ZX diagrams, circuits, and tableaux	Maarten Grothuis, V. Vilasini Impossibility of superluminal signalling rules out causal loops in conical spacetimes
Boat tour 17:30 – 18:30			
Conference dinner starting 19:00			

Thursday, August 20th

Long plenary talks	
9:30 – 10:15	Cole Comfort, Robert I. Booth Denotational semantics for stabiliser quantum programs
10:15 – 11:00	Kathleen Barse, Romain Péchoux, Simon Perdrix Quantum control and general recursion beyond the unitary case
Coffee break	
Short plenary talk	
11:30 – 11:55	Aabhas Gulati, Ion Nechita, Clément Pellegrini Entanglement in the Dicke subspace
Conference photo 11:55 – 12:20	
Lunch break	

	Parallel A	Parallel B	Parallel C
14:30 – 14:55	Arianne Meijer-van de Griend, Leo Becker Pauli gadget synthesis for gatesets with arbitrary even-arity Clifford gates	Haruki Emori Quantum statistical functions	Nadish de Silva, Santanil Jana, Ming Yin Three-qubit nonlocality paradoxes: beyond GHZ
14:55 – 15:20	Soichiro Yamazaki, Seiseki Akibue Multi-qubit controlled gate with optimal T-count	Michael Zurel, Jack Davis Basis-independent stabilizerness and maximally noisy magic states	Shashaank Khanna, Matthew Pusey, Roger Colbeck Identifying causal structures which cannot support quantum correlations without fine-tuning
15:20 – 15:45	Akash Kundu Tensor and gadget reinforcement learning for improved, hardware-aware quantum architecture search	Cameron Calcluth, Oliver Hahn, Juani Bermejo Vega, Alessandro Ferraro, Giulia Ferrini Classical simulation of circuits with realistic odd-dimensional Gottesman–Kitaev–Preskill states	C. E. Lopetegui-Gonzalez, G. Masse, E. Oudot, U. I. Meyer, F. Centrone, F. Grosshans, P. E. Emeriau, U. Chabaud, M. Walschaers A unified framework for Bell inequalities from continuous-variable contextuality
15:45 – 16:10	Mark Deaconu, Nihar Gargava, Amolak Ratan Kalra, Michele Mosca, Jon Yard Buildings for synthesis with Clifford+R	Massimo Frigerio, Mattia Walschaers, Andrei Aralov, Carlos Ernesto Lopetegui-Gonzalez, Emilie Gillet Algebraic techniques for photonic state preparation and characterization	Martin J. Renner, Edwin Peter Lobo, Arturo Konderak, Remigiusz Augusiak, Antonio Acín Full nonlocality for non-maximally entangled states
Coffee break 16:10 – 16:40			
Career fair 16:40 – 19:00			

Friday, August 21st

Long plenary talks	
9:30 – 10:15	Thea Li, Vladimir Zamdzhiev Quantum coherence spaces revisited: a von Neumann (co)algebraic approach
10:15 – 11:00	Matilde Baroni, Dominik Leichtle, Ivan Šupić, Damian Markham, Marco Túlio Quintino Composable simultaneous purification: when all communication scenarios reduce to spatial correlations
Coffee break	
Business meeting 11:30 – 12:20	
Lunch break	

	Parallel A	Parallel B	Parallel C
14:30 – 14:55	Simon Burton, Hussain Anwar Meromorphic quantum computing	Noé Delorme, Simon Perdrix Diagrammatic reasoning with control as a constructor, applications to quantum circuits	David Schmid, John H. Selby, Vinicius Pretti Rossi, Roberto D. Baldijão, Ana Belén Sainz Shadows and subsystems of generalised probabilistic theories: when tomographic incompleteness is not a loophole for contextuality proofs
14:55 – 15:20	Christine Li, Lia Yeh Transversal AND in quantum codes	Chris Heunen, Robin Kaarsgaard, Louis Lemonnier One rig to control them all	Jan-Åke Larsson The contextual Heisenberg microscope
15:20 – 15:45	Jin Ming Koh, Anqi Gong, Andrei C. Diaconu, Daniel Bochen Tan, Alexandra A. Geim, Michael J. Gullans, Norman Y. Yao, Mikhail D. Lukin, Shayan Majidy Phantom codes: entangling logical qubits without physical operations	William Schober, Scott Wesley A complete equational theory for quantum circuits with generalized control	Daniel McNulty Quantifying quantum measurement incompatibility via graph invariants
Coffee break			
16:15 – 16:40	Vivien Vandaele Asymptotically optimal quantum circuits for comparators and incrementers	Sacha Cerf, Harold Ollivier The perturbative method for quantum correlations	Raffaele D'Avino, Lorenzo Caramelli, Raja Yehia, Gabriel Senno, Roberto González Pousa, Antonio Acín, Tamás Kriváchy Device independent quantum key distribution with a single measurement per site
16:40 – 17:05	Giuseppe De Riso, Giuseppe Catalano, Seth Lloyd, Vittorio Giovannetti, Dario De Santis A resource-efficient quantum-walker quantum RAM	Subhendu Bikas Ghosh, Snehasish Roy Chowdhury, Guruprasad Kar, Arup Roy, Tamal Guha, Manik Banik Strong inequivalence of quantum nonlocal resources	Paul Becsi, Matthew Joseph Hoban Bounding classical and quantum correlations in Bayesian networks with quasiprobabilities
End of QPL 2026. Goodbye!			